

HW06 - API Testing Report

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Repository: <https://github.com/linhnph05/QA-HW06>

Self-assessed score: 100/100

1. Scope

I tested three EShop features:

Pool	Feature	Selected endpoints
A	FR-03 Forgot and reset password	POST /api/forgot-password, POST /api/reset-password
B	FR-09 Discount coupons	POST /api/apply-coupon, POST /api/coupon-usage
C	FR-12 Access control	GET /api/admin/users, DELETE /api/admin/users/:id, GET /api/admin/orders, PUT /api/admin/orders/:id/status, GET /api/coupons, POST /api/admin/coupons, DELETE /api/admin/coupons/:id, POST /api/admin/import-products, POST /api/products

I used the API specification, security rules SEC-01 to SEC-07, and the running backend at <http://127.0.0.1:3000>.

2. AI-First Test Design and Human Audit

I guided three helper AI sessions separately. Each helper received only the relevant feature, endpoints, rules, and required test design areas. Each produced exactly 35 draft cases. I then checked every case against the specification and code.

Feature	AI cases	Valid	Invalid	Incomplete	Human additions	Final cases
FR-03	35	30	1	4	5	40
FR-09	35	29	1	5	5	40
FR-12	35	30	1	4	5	40
Total	105	89	3	13	15	120

The detailed generation, audit, and extension files are in test-cases/. The exact helper prompts and outputs are attached through AI-AUDIT-REPORT.md.

The human additions focus on cases that AI often misses: account enumeration, rate limits, concurrent OTP use, coupon usage races, numeric overflow, known-secret JWT forgery, stale roles, deleted users, and admin self-delete.

3. Execution

I converted the 120 reviewed cases to one strict Postman collection and ran it with Newman 6.2.2. I also created a separate FR-03 data-driven collection whose request values and expected results come from a CSV file. A collection pre-request script adds X-Student-Id: 23127081 to every request.

Commands and Postman steps used

I started the local backend with:

```
cd eshop-sut/backend
node server.js
```

Postman desktop did not use a shell command. I used these steps in the application:

1. Click Import and import collections/EShop-HW06.postman_collection.json.
2. Import collections/local.postman_environment.json and select the local environment.
3. Open the collection runner and run the FR-03, FR-09, and FR-12 folders against http://127.0.0.1:3000.
4. For the data-driven demonstration, select EShop-HW06-data-driven.postman_collection.json and load data/fr03-runner-data.csv as the Runner data file.
5. Check the Postman Console to confirm X-Student-Id: 23127081 on each request.

I ran the strict local suite and exported the JSON and HTML reports with:

```
newman run collections/EShop-HW06.postman_collection.json \
-e collections/local.postman_environment.json \
-r cli,json,html \
--reporter-json-export newman/newman-results.json \
--reporter-html-export newman/newman-report.html
```

I ran the real CSV iteration-data demonstration with:

```
newman run collections/EShop-HW06-data-driven.postman_collection.json \
-e collections/local.postman_environment.json \
-d data/fr03-runner-data.csv \
-r cli,json,html \
--reporter-json-export newman/data-driven-results.json \
--reporter-html-export newman/data-driven-report.html
```

The data file supplied four iterations. Newman executed four requests and all 16 assertions passed. The collection reads case_id, email, expected_status, and schema_kind through pm.iterationData.

Newman Report

Collection
Description
Time
Exported with

EShop HW06 Data-Driven Demo - 23127081
A real Newman iteration-data run for FR-03. Each CSV row supplies the email, expected status, schema kind, and case ID.
Sat Aug 29 2025 19:07:19 GMT+0700 (Indochina Time)
Newman v6.2.2

	Total	Failed
Iterations	4	0
Requests	4	0
Prerequisite Scripts	4	0
Test Scripts	4	0
Assertions	16	0
Total run duration		240ms
Total data received		218B (approx)
Average response time		17ms
Total Failures	0	

Requests

FR-03 forgot-password from CSV row

Method	POST		
URL	http://127.0.0.1:3000/api/forgot-password		
Mean time per request	17ms		
Mean size per request	55B		
Total passed tests	16		
Total failed tests	0		
Status code	200		
Tests	Name	Pass count	Fail count
	DD-FR03-01 - status	1	0
	DD-FR03-01 - student header	1	0
	DD-FR03-01 - JSON content type	1	0
	DD-FR03-01 - exact response schema	1	0
	DD-FR03-02 - status	1	0
	DD-FR03-02 - student header	1	0
	DD-FR03-02 - JSON content type	1	0
	DD-FR03-02 - exact response schema	1	0
	DD-FR03-03 - status	1	0
	DD-FR03-03 - student header	1	0
	DD-FR03-03 - JSON content type	1	0
	DD-FR03-03 - exact response schema	1	0
	DD-FR03-04 - status	1	0
	DD-FR03-04 - student header	1	0
	DD-FR03-04 - JSON content type	1	0
	DD-FR03-04 - exact response schema	1	0

Data-driven Newman run with four CSV iterations

GitHub Actions ran the passing characterization baseline with:

```
newman run collections/EShop-HW06-ci-baseline.postman_collection.json \
-e collections/local.postman_environment.json \
```

```
-r cli,json \
--reporter-json-export ci-results/newman-ci.json \
--color off
```

Feature	Executed	Passed	Failed	Failed assertions
FR-03	40	17	23	25
FR-09	40	20	20	26
FR-12	40	17	23	23
Total	120	54	66	74

There were 389 assertions: 315 passed and 74 failed. The collection-level test script uses `pm.response.to.have.jsonSchema` for every selected test request. Each mapped schema checks the exact allowed keys, required keys, field types, array item types, and `additionalProperties: false`. Error responses must contain only a string error field. All schema assertions passed; the remaining failures show security, validation, or business-logic mismatches. Evidence is in `newman/newman-report.html`, `newman/newman-results.json`, `results/*.csv`, `images/newman-summary.png`, and `images/student-id-console.png`. The console screenshot proves that the pre-request script added `X-Student-Id: 23127081` to the executed requests.

Newman Report

Collection

Description

Time

Exported with

EShop HW06 API Tests - 23127081

120 audited API cases for FR-03, FR-09, and FR-12.

Sat Aug 29 2026 21:30:16 GMT+0700 (Indochina Time)

Newman v6.2.2

	Total	Failed
Iterations	1	0
Requests	140	0
Prerequisite Scripts	140	0
Test Scripts	248	0
Assertions	389	74

Total run duration

Total data received

Average response time

2.6s

16.71KB (approx)

2ms

Total Failures

74

Requests

Setup

Register runner user

Method

URL

Mean time per request

Mean size per request

Total passed tests

Total failed tests

Status code

POST

<http://127.0.0.1:3000/api/register>

39ms

49B

0

0

200

Register reset user

Method

URL

Mean time per request

Mean size per request

POST

<http://127.0.0.1:3000/api/register>

4ms

49B

Newman execution summary

```

    'X-Student-Id header:', '23127081'
  POST http://127.0.0.1:3000/api/apply-coupon [404 Not Found, 353B, 4ms]
56. FR09-AI-030 - status is 401
✓ FR09-AI-030 - response is JSON

↳ FR09-AI-031 - domain/security/schema
    'X-Student-Id header:', '23127081'
  POST http://127.0.0.1:3000/api/apply-coupon [404 Not Found, 353B, 3ms]
57. FR09-AI-031 - status is 403
✓ FR09-AI-031 - response is JSON

↳ FR09-AI-032 - domain/security/schema
    'X-Student-Id header:', '23127081'
  POST http://127.0.0.1:3000/api/apply-coupon [404 Not Found, 353B, 3ms]
58. FR09-AI-032 - status is 200
✓ FR09-AI-032 - response is JSON

↳ FR09-AI-033 - domain/security/schema
    'X-Student-Id header:', '23127081'
  POST http://127.0.0.1:3000/api/apply-coupon [404 Not Found, 353B, 3ms]

```

X-Student-Id header in the Newman console

4. Postman and Newman Features Used

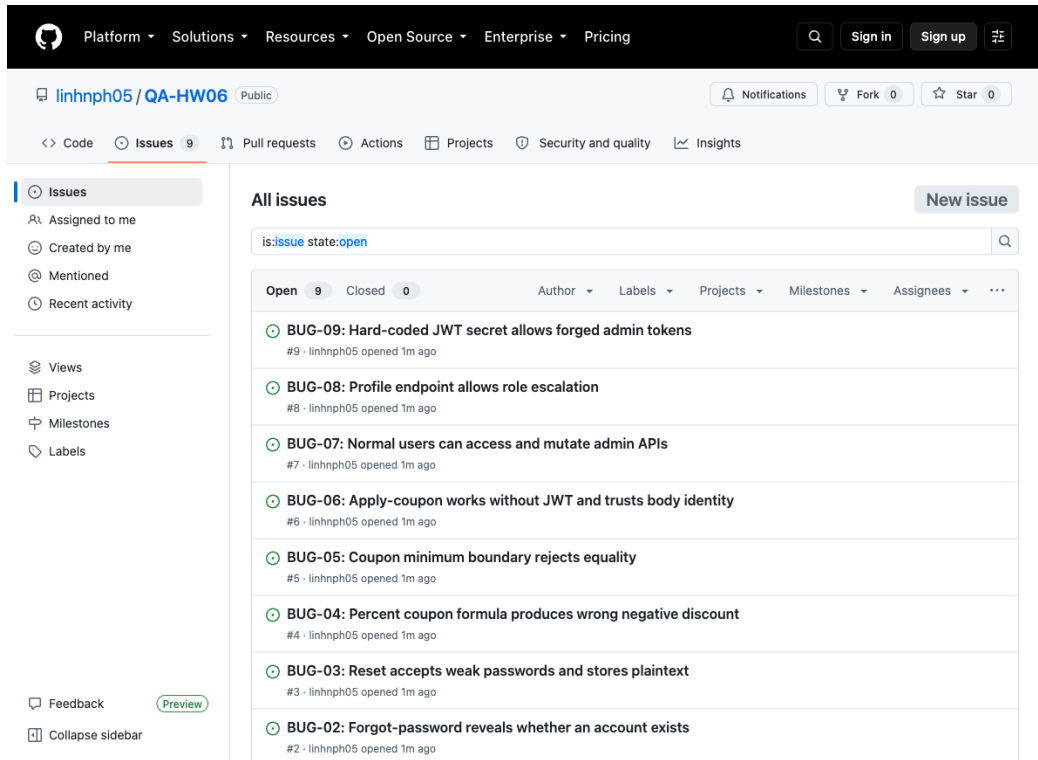
- Collection folders for the three selected features.
- Collection variables and an environment file for the base URL, accounts, tokens, OTPs, IDs, and coupon data.
- A collection-level pre-request script for the student header.
- Request-level test scripts and dynamic state sharing.
- A separate data-driven collection using `pm.iterationData`, the Collection Runner data-file input, and Newman's `-d data/fr03-runner-data.csv` option.
- Positive, negative, boundary, state, and security assertions, plus endpoint-specific exact JSON Schema checks using `pm.response.to.have.jsonSchema`.
- Newman CLI execution with CLI, JSON, and HTML reporters.
- GitHub Actions execution and uploaded JSON result artifact.

5. Bugs

I reported nine genuine bugs in both `reports/bug-report.md` and GitHub Issues:

1. Four-digit OTP with no expiry state: <https://github.com/linhnph05/QA-HW06/issues/1>
2. Forgot-password account enumeration: <https://github.com/linhnph05/QA-HW06/issues/2>
3. Weak and plaintext reset password: <https://github.com/linhnph05/QA-HW06/issues/3>
4. Wrong percent coupon formula: <https://github.com/linhnph05/QA-HW06/issues/4>
5. Wrong minimum-total boundary: <https://github.com/linhnph05/QA-HW06/issues/5>
6. Coupon API lacks JWT enforcement: <https://github.com/linhnph05/QA-HW06/issues/6>
7. Admin role is not enforced: <https://github.com/linhnph05/QA-HW06/issues/7>
8. Profile update allows role escalation: <https://github.com/linhnph05/QA-HW06/issues/8>
9. Hard-coded JWT secret permits forgery: <https://github.com/linhnph05/QA-HW06/issues/9>

Every issue includes its own screenshot. The Issues-page evidence is `images/github-issues.png`.



GitHub Issues page with the nine reported bugs

6. CI/CD

The workflow in `.github/workflows/api-tests.yml` checks out the SUT submodule, installs Node.js and Newman, starts the backend, runs all 120 baseline cases, and uploads the Newman JSON artifact.

- All-passing commit: 98f77c3aa64d70662ba6aa3b4d4cfad9fed421e8
- All-passing run: <https://github.com/linhnph05/QA-HW06/actions/runs/32109080739>
- One-failure commit: b6e490c6de73f3ae811cb0a0f6e0bab75b7d93ef
- One-failure run: <https://github.com/linhnph05/QA-HW06/actions/runs/32109235988>
- Final restored green run: <https://github.com/linhnph05/QA-HW06/actions/runs/32109618742>

The red sample changes only one expected status from 200 to 418. It produces exactly one failed assertion. I restored it afterward. More detail is in `reports/ci-cd-report.md`.

The screenshot shows a GitHub Actions workflow run for the repository 'linhnph05 / QA-HW06'. The workflow is named 'CI: configure EShop submodule checkout #2' and is currently in a 'Success' state. It was triggered by a push to the 'main' branch by user 'linhnph05' 1 minute ago. The total duration of the run is 25 seconds, and there is 1 artifact. The workflow file is 'api-tests.yml' and it runs on a 'push' event. The job 'newman' completed successfully in 22 seconds. There is 1 warning annotation: 'newman: Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js ...'. The artifacts section is empty.

All-passing GitHub Actions run

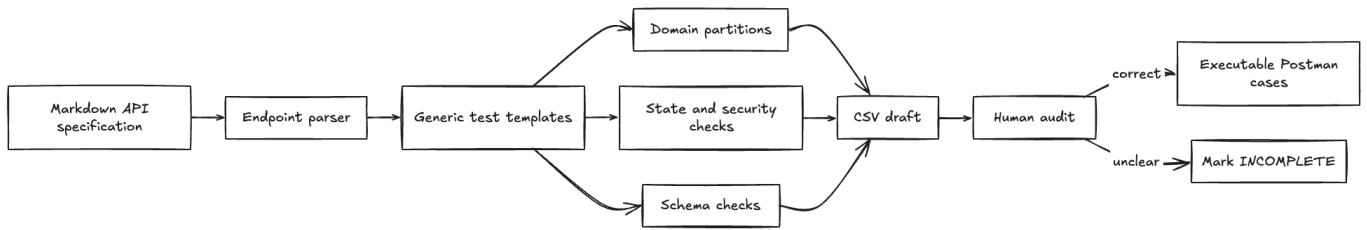
The screenshot shows a GitHub Actions workflow run for the repository 'linhnph05 / QA-HW06'. The workflow is named 'CI: demonstrate exactly one failing API test #3' and is currently in a 'Failure' state. It was triggered by a push to the 'main' branch by user 'linhnph05' 3 minutes ago. The total duration of the run is 26 seconds, and there is 1 artifact. The workflow file is 'api-tests.yml' and it runs on a 'push' event. The job 'newman' failed in 22 seconds. There are 2 annotations: 1 error and 1 warning. The error is 'newman: Process completed with exit code 1.'. The warning is 'newman: Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js ...'. The artifacts section shows 1 artifact: 'newman-ci-result' with a size of 71.1 KB and a digest of 'sha256:77cd7df83ba9a31...'. The artifact is produced during runtime.

GitHub Actions run with exactly one failed assertion

7. AI-Driven Test Generator Skill

I created the reusable api-test-generator Agent Skill. It reads Markdown endpoint definitions and writes a CSV test draft. The draft includes domain, boundary, security, header, and schema templates. All generated rows start as INCOMPLETE so a human must audit them. The skill validator passed, and the FR-03 demonstration produced test-cases/fr03-skill-demo.csv.

I drew the generator diagram using excalidraw. The PNG and pseudocode are in images/api-test-generator-diagram.png and reports/api-test-generator-design.md.



Student-drawn API test generator diagram

8. AI Critique

The full 200–300 word critique is in `reports/ai-critique.md`. My main lesson is that AI is useful for broad coverage, but a human must challenge its assumptions and connect separate security facts.

9. AI Audit Appendix

I use AI tools for the tasks declared in `AI-AUDIT-REPORT.md`. That appendix contains only my interactions with the three helper AIs. It does not include the conversation with the assistant that helped manage this repository.

10. Self-Assessment

No.	Criteria	Grade	Self-Assessed Grade
1	FR-03 full pipeline	30	30
2	FR-09 full pipeline	30	30
3	FR-12 full pipeline	30	30
4	Agent Skill	10	10
	Total	100	100/100